

the plane, destroying the defenceless target and its civilian occupants with horrific precision.

While conspiracy stories blame the then active Bush administration for carrying out what they call a 'cold blooded massacre' as a show of aggression, one thing is for certain, the captain of USS Vincennes was given faulty and misleading information as a result of very poor UI design for the Aegis software.

Not only was the craft ascending (and not descending, as it was reported to be) but the craft's Identification, Friend or Foe (or IFF) reading, designed to distinguish between civilian and military craft, was reported to be Mode II (military) and not Mode III (civilian, as it actually was).

This mistake cost the lives of all the passengers and crew on board the Iranian flight and was the cause of much public humiliation for the United States on the world stage.

Scud missile hits American barracks undetected by US air defences(1991):

Twenty-eight Americans were killed on February 25, 1991 when a Scud missile hit the Army barracks in Dhahran, Saudi Arabia. The Patriot defence system had failed to track and intercept the missile.

The Patriot missile saves time as a fixed point register that had a length of 24 bits. Since the internal clock of the system is measured every one-tenth of a second, $1/10$ expressed in a 24 bit fixed point register is



Faulty software can lead to terrible consequences